

SOC ANALYST LEVEL 1 COMPLETE FREE ROADMAP — 2026

BCA Sem 2 → Job Ready | India → Estonia

Candidate	Shivam Kumar — BCA Student (Semester 2)
Goal 1	SOC Analyst Level 1 — India — by BCA Final Year
Goal 2	Security Analyst Level 2 — Estonia/EU (Long Term)
Strategy	100% VERIFIED FREE Resources ONLY — Zero paid courses
Version	FINAL v2 — June 2026 — TryHackMe removed, all resources re-verified

IMPORTANT: Is version mein TryHackMe ko hata diya gaya hai kyunki unki zyada tar rooms ab paid ho chuki hain. Yahan har ek resource personally verify kiya gaya hai — sab 100% free hain. Yeh aapka FINAL roadmap hai — dobara mat badlo.

AAPKI BCA TIMELINE — Semester-Wise Map

SEMESTER	PHASE	FOCUS	TARGET
Sem 2 (NOW)	Phase 1 Start	Networking + Linux + Windows Basics	Foundation solid — notes ready
Sem 3	Phase 1 Complete	Cybersecurity fundamentals + Command line	Cisco badge + IBM badge earned
Sem 4	Phase 2 Start	SIEM + Log Analysis + Wireshark + Sysmon	Splunk + Wazuh setup done, GitHub started
Sem 5	Phase 2 Core	EDR + AD + Threat Frameworks + KQL/SPL	15 GitHub writeups + CyberDefenders done
Sem 6 Final	Phase 3 Job Prep	Portfolio complete + Interview prep + Applying	Resume ready + 25+ writeups + Applications sent
After BCA +3 months	Phase 4	SOC Internship / Junior Analyst Role	First SOC job secured

PHASE 1 — Foundation Building | Sem 2 → Sem 3

Yahan se shuru karo. Yeh skills ke bina SOC ka koi bhi alert samajh nahi aayega.

TOPIC	KYA SEEKHNA HAI
OSI Model	7 layers — Layer 1 se 7 tak. Har layer ka kaam. Real attacks kahan hote hain.
TCP/IP	3-way handshake, SYN/ACK/RST/FIN flags, packets, IP headers padhna
DNS	Resolution kaise hoti hai, A/MX/CNAME records, DNS poisoning concept
HTTP/HTTPS	GET/POST requests, status codes 200/301/403/404/500, TLS handshake
Common Ports	21-FTP, 22-SSH, 23-Telnet, 25-SMTP, 53-DNS, 80-HTTP, 443-HTTPS, 445-SMB, 3389-RDP
IP + Subnetting	IPv4/IPv6, CIDR (/24 /16), private vs public IPs, NAT concept
Linux Commands	cd, ls, grep, find, cat, chmod, chown, ps, netstat, curl — roz practice karo
Linux Log Files	/var/log/auth.log, /var/log/syslog — locations yaad rakho
Linux Permissions	rwX model, chmod numerics 755/644, sudo basics
Windows Basics	Users, Groups, Services, Task Manager, Registry basics, C:\Windows\System32
Windows Event Viewer	Security/System/Application log categories. Events filter karna.
PowerShell Basics	Get-Process, Get-Service, Get-EventLog. PS script padhna aur samajhna.
Command Line	CMD + Linux Terminal — terminal se bilkul darr nahi hona chahiye

Phase 1 — 100% FREE Resources (Verified)

PLATFORM	KYA KARNA HAI	URL	COST
Cisco Skills For All	Networking Basics (complete) + Junior Cybersecurity Analyst Career Path. ~120 hours. Digital badge milta hai LinkedIn ke liye.	skillsforall.cisco.com	FREE
LinuxJourney	Linux ka best free interactive tutorial. Grasshopper se Journeyman tak sab complete karo. Browser mein kaam karta hai.	linuxjourney.com	FREE
OverTheWire Bandit	Linux hands-on practice — game format mein. Level 0 se Level 20 tak karo. Real terminal use hoti hai. Best Linux practice platform.	overthewire.org/wargames/bandit	FREE
Professor Messer YouTube	CompTIA Network+ N10-008 full video series — networking ka sabse acha free explanation. Cisco ke saath saath watch karo.	YouTube: 'Professor Messer Network+'	FREE
Microsoft Learn	Windows fundamentals, PowerShell basics — Microsoft ka official free content. SC-900 bhi yahan se hai.	learn.microsoft.com	FREE
IBM SkillsBuild	Cybersecurity Fundamentals course — IBM digital badge milta hai. LinkedIn pe add karo.	skillsbuild.org	FREE
subnetting.net	Sirf subnetting practice ke liye. Interactive tool — tab tak practice karo jab tak second nature na ban jaye.	subnetting.net	FREE
Fortinet NSE 1 & 2	Threat landscape aur cybersecurity basics — completely free certificate milta hai. 10-15 hours ka course.	training.fortinet.com	FREE

Sem 2-3 Target: Cisco Networking Basics + LinuxJourney complete karo. OverTheWire Bandit Level 20 tak pahuncho. GitHub account banao aur pehla repo 'Learning-Notes' create karo.

PHASE 2 — SOC Core Skills | Sem 4 → Sem 5

Yeh roadmap ka sabse important phase hai. Yahan actual SOC skills build hoti hain. Ek ek topic dhyan se karo.

2A. SIEM — Kam se Kam 2 Platforms Seekho

TOOL	KYA SEEKHNA HAI	FREE RESOURCE	CO ST
Splunk (Primary)	SPL queries: search stats eval rex. Dashboards, alert creation, log correlation. 40+ hours practice.	splunk.com/en_us/training/free-courses.html — Splunk Fundamentals 1 free course	FRE E
Microsoft Sentinel	KQL queries basics, analytics rules, cloud SIEM architecture. SC-900 ke saath seekho.	learn.microsoft.com — SC-900 + SC-200 free modules	FRE E
Wazuh (EDR+SIEM)	Open-source SIEM+EDR. Agent setup, rule alerting, MITRE mapping. Install karo aur use karo.	wazuh.com — free download + docs. YouTube: 'Install Wazuh step by step'	FRE E
ELK Stack	Kibana dashboard, basic queries. Bahut Indian SOCs use karte hain.	elastic.co free cloud trial — 14 days kaafi hai basics ke liye	FRE E trial

2B. Essential SOC Tools

TOOL	KYA KARTA HAI	FREE LINK	CO ST
Wireshark	Packet capture, network traffic analysis, PCAP files padhna	wireshark.org	FRE E
Sysmon	Advanced Windows logging — process, network, registry changes	Microsoft Sysinternals (free download)	FRE E
VirusTotal	File hash / URL / IP / domain reputation check — IOC analysis	virustotal.com	FRE E
Any.Run	Interactive malware sandbox — file run karo safely	any.run (free community tier)	FRE E
CyberChef	Base64, hex, encoding/decoding — log manipulation	gchq.github.io/CyberChef	FRE E
TheHive	Incident case management — alerts track karo, document karo	thehive-project.org	FRE E
URLScan.io	Suspicious URLs scan karo bina visit kiye	urlscan.io	FRE E
MxToolbox	Email header analysis, SPF/DKIM/DMARC check	mxtoolbox.com	FRE E
MISP	Threat Intelligence — IOC management, threat feeds	misp-project.org	FRE E
Abuse.ch / Bazaar	Malware hash lookup, botnet C2 tracking	bazaar.abuse.ch	FRE E
Windows Event Viewer	Native Windows log analysis — Security/System/Application	Built-in Windows tool	FRE E
Regex101	Regex practice — log se data extract karna practice	regex101.com	FRE E

PHASE 2 (Jari) — Event IDs + Frameworks + Identity

2C. Windows Event IDs — 15 Must-Know

ID	MATLAB	SOC MEIN KYU	LEVEL
4624	Successful Logon	Normal logins baseline karo	Monitor
4625	Failed Logon	Brute force / password spray	HIGH
4648	Explicit Credentials Logon	Pass-the-hash / lateral movement	HIGH
4672	Special Privileges Assigned	Admin account activity track karo	HIGH
4688	Process Creation	Malware / suspicious process — parent chain dekho	HIGH
4698	Scheduled Task Created	Attacker persistence — common tactic	HIGH
4720	User Account Created	Unauthorized backdoor account	HIGH
4726	User Account Deleted	Attack ke baad cleanup	Monitor
4732	User Added to Security Group	Privilege escalation	CRITICAL
4740	Account Locked Out	Brute force confirmation	HIGH
4771	Kerberos Pre-Auth Failed	Kerberoasting indicator	HIGH
4776	NTLM Auth Attempt	Pass-the-hash indicator	HIGH
1102	Audit Log Cleared	CRITICAL — attacker tracks cover kar raha hai	CRITICAL
7045	New Service Installed	Malware persistence via service	HIGH
4104	PowerShell Script Block	Malicious PS commands detect karo	HIGH

2D. Threat Frameworks — Dono Seekhna Zaroori

FRAMEWORK	KYA HAI	KYA SEEKHNA HAI	FREE RESOURCE	CO ST
MITRE ATT&CK;	Attack tactics & techniques database	14 tactics, top 30 techniques, ATT&CK; Navigator use karna	attack.mitre.org	FRE E
Cyber Kill Chain	7-stage attack model	7 stages: Recon→Weapon→Deliver→Exploit→Install→C2→Action	lockheedmartin.com free PDF	FRE E
NIST IR	Industry IR standard	6 phases: Prep→Detect→Contain→Eradicate→Recover→Lessons	nvlpubs.nist.gov NIST SP 800-61 free PDF	FRE E
Diamond Model	Threat modeling	Adversary, Capability, Infrastructure, Victim — concept	Google 'Diamond Model cybersecurity PDF'	FRE E

2E. Active Directory & Identity — Bahut Important

- Active Directory kya hai — Domain, Forest, OU, Domain Controller
- Users aur Groups — Security Groups, permission inheritance
- Kerberos — TGT, TGS tickets ka basic flow
- Common AD attacks naam yaad karo — Pass-the-Hash, Kerberoasting, DCSync
- Azure AD / Entra ID — cloud identity basics
- Key Event IDs: 4771, 4776, 4732, 4648

FREE Resource: Microsoft Learn — Active Directory fundamentals (free) | Google 'Active Directory basics for beginners' — bahut free YouTube content available

PHASE 2 (Jari) — Investigations + KQL/SPL + Free Practice Platforms

2F. SOC Investigation Steps

Phishing Investigation

- Email header check — SPF/DKIM/DMARC
- Sender domain — whois se check karo
- Links — URLScan.io se scan karo
- Attachment hash — VirusTotal pe check
- Sandbox — Any.Run mein run karo
- IOCs extract — IP, domain, hash, URL
- MITRE ATT&CK; se map karo
- Case management mein document karo

Malware Alert Investigation

- Process name, parent process, command line padho
- Process tree dekho — chain samjho
- File hash VirusTotal pe check karo
- Network connections dekho — kahan ja raha hai
- Sysmon Event ID 1, 3, 13 correlate karo
- MITRE technique identify karo
- Timeline banao — pehla event kab hua
- Escalate ya close — reason document karo

2G. KQL + SPL Query Languages

LANGUAGE	PLATFORM	EXAMPLE QUERY	FREE PRACTICE
SPL	Splunk	index=main search failed stats count by src_ip where count > 5 sort -count	Splunk Free Training
KQL	Sentinel	SecurityEvent where EventID == 4625 summarize count() by Account where count_ > 10	Microsoft Learn free

Phase 2 — Verified Free Practice Platforms

PLATFORM	KYA FREE MILTA HAI	URL	COST
LetsDefend	15 SOC alerts/month investigate karo free mein. Free courses: SOC Fundamentals, Phishing Analysis, Linux Blue Team, Splunk.	letsdefend.io	FREE
CyberDefenders	PCAP analysis, forensics, threat hunting challenges — sab free. Real investigation practice. Best blue team platform.	cyberdefenders.org	FREE
Blue Team Labs Online	Weekly challenges free tier mein — Splunk, Windows forensics, log analysis scenarios.	blueteamlabs.online	FREE tier
Splunk Training	Splunk Fundamentals 1 — official free SPL course. Certificate milta hai.	splunk.com/en_us/training/free-courses.html	FREE
Microsoft Learn	SC-900, AZ-900, SC-200 modules, Sentinel free training. Sab free.	learn.microsoft.com	FREE
Google Coursera	Google Cybersecurity Certificate — Financial Aid apply karo. Approved hone par bilkul free. 8 strong modules.	coursera.org — Financial Aid	FREE (Aid se)
EC-Council Essentials	SOC Essentials free series — SOC architecture, SIEM basics, log management overview.	eccouncil.org free courses	FREE
Fortinet NSE 1-3	Free cybersecurity certificates. NSE 1 & 2 completely free, NSE 3 bhi free.	training.fortinet.com	FREE
SANS Reading Room	Free white papers — SOC operations, IR, threat hunting. Deep technical reading.	sans.org/reading-room	FREE

The DFIR Report	Real intrusion case studies — attacker TTPs samajhne ke liye mahine mein ek padhna.	thedfirreport.com	FRE E
Sigma HQ GitHub	3000+ detection rules — detection logic samajhne ke liye padhna.	github.com/SigmaHQ/sigma	FRE E
Wazuh Docs	Complete SIEM+EDR documentation + installation guides.	wazuh.com/documentation	FRE E

3A. GitHub Portfolio Structure

- **Repo Name:** SOC-Portfolio ya CyberSecurity-Journey
- **Folders:** 01-Networking-Notes / 02-Linux-Notes / 03-Log-Analysis / 04-Investigations / 05-SIEM-Queries / 06-Incident-Reports
- **Har lab ke baad:** README.md mein likho — Scenario → Tools → Steps → Findings → IOCs → MITRE Technique → Lesson
- **Screenshots:** Terminal output, VirusTotal results, SIEM queries — sab add karo
- **Minimum:** 1 writeup per week. 25+ writeups by final semester.
- **Pinned repos:** Best 4-5 repos GitHub profile pe pin karo

3B. LinkedIn Strategy

- **Headline:** 'Aspiring SOC Analyst | Blue Team | Cybersecurity Student'
- **About:** 2-3 lines — goal, skills, platforms, kya seekh rahe ho. Simple aur honest.
- **Weekly:** Ek lab writeup ya learning share karo.
- **Badges add karo:** Cisco, IBM SkillsBuild, Splunk, Fortinet NSE, SC-900
- **Connect:** Indian SOC professionals, HR at Paladion, HCL Cyber, Wipro CyberSec
- **Avoid:** Fake certifications, exaggerated claims — sab verify hota hai interviews mein

3C. Communication Practice — Daily

ACTIVITY	KAISE PRACTICE KARO	TIME	COST
Technical Writing	Har lab ke baad 5 complete English sentences mein summary likho. Bullets nahi — full sentences.	15 min daily	FREE
English Listening	Darknet Diaries podcast — 1 episode per week. Real security stories. darknetdiaries.com	Weekly	FREE
Mock Interview	Phone pe record karo — 'Apni ek investigation explain karo.' Watch back aur improve karo.	Weekly	FREE
Alert Explanation	'Mujhe yeh alert mila tha...' — story format mein explain karna practice karo.	5 min daily	FREE

PHASE 4 — Job Application | Final Sem + 3 Months After BCA

COMPANY TYPE	EXAMPLES	SEARCH TERM
MSSPs — Freshers ke liye best	Paladion Networks, SAFE Security (Lucideus), TAC Security, Sequaretek, SISA, Aujas	'SOC Analyst Fresher MSSP India'
Large IT Companies	HCL Tech CyberSec, Wipro CyberSec, Infosys Cyber, TCS CyberSec, Tech Mahindra	'Junior SOC Analyst HCL Wipro'
Product Security Companies	Quick Heal, K7 Security, Seqrite, McAfee India, Trend Micro India	'Security Analyst Quick Heal K7'
SOC Internships — Pehle yeh	Upar wali koi bhi company + startups + CERT-In partner companies	'SOC Analyst Internship India'
Job Boards	LinkedIn Jobs, Naukri.com, Indeed India — daily check karo, alerts set karo	Job alerts set karo — 'SOC Fresher'

Interview Mein Jo Poochha Jaata Hai

CATEGORY	QUESTIONS	PREPARATION
SOC Basics	SOC kya hai? Tier 1 vs 2? Alert triage process?	Apne words mein explain karna practice karo
Technical Tools	Splunk mein failed logins kaise dekhte ho? Wireshark mein SYN scan?	Actual tools use karo — theory se kaam nahi chalta
Scenario Based	'4625 events zyada aa rahe hain — kya karoge?' 'Phishing email — steps?'	Investigation steps samjho, memorize nahi karo
Frameworks	MITRE ATT&CK; kya hai? Kill Chain stages? T1566?	attack.mitre.org explore karo
Behavioral	Apna best investigation describe karo.	GitHub writeup ready rakho — real example se answer karo

Cloud SOC + AI + Estonia — Long-Term Sections

Cloud SOC Basics (2026)

- SC-900: Microsoft Security Fundamentals — FREE on learn.microsoft.com. Complete karo.
- Microsoft Sentinel — KQL, analytics rules, cloud SIEM
- Azure logs, Azure Monitor, Log Analytics workspace basics
- Cloud identity attacks — MFA bypass, service account compromise
- AWS CloudTrail — awareness only at L1

AI Security (2026 Awareness)

- Prompt Injection — AI models manipulate karna
- AI-generated phishing — perfect English, highly targeted
- AI-assisted log analysis — speed up SOC work
- ALWAYS manually verify AI outputs before escalating
- L1 ke liye basic awareness kaafi hai

AREA	KYA KARNA HAI	FREE RESOURCE	TIMELINE
SOC Experience	Minimum 3-4 saal India mein genuine SOC experience. Bina iske Estonia apply karna time waste.	—	After BCA +3-4 years
GDPR	European data privacy law — personal data, breach reporting, controller vs processor	gdpr.eu (official free)	Year 2-3 of job
ISO 27001	Information security management — concept awareness, certification zaruri nahi	Google 'ISO 27001 intro PDF'	Year 2-3
English	Professional-level English — Estonia sab English mein hai	Darknet Diaries + BBC Learning English (free)	Start now
CySA+	Best L2 certification EU employers ke liye. 2 saal experience ke baad start karo.	Professor Messer YouTube free prep	2-3 years into job

Daily Routine + Phase 5 (After Job)

TIME	ACTIVITY	DURATION	NOTE
Subah / College	BCA studies — degree sabse pehle. Fail mat karo cybersecurity ke chakkar mein.	3-4 hours	Non-negotiable
Dopahar	Cybersecurity theory — notes padhna, 1 YouTube video, concept revision	1 hour	Daily karo
Shaam	Hands-on Labs — CyberDefenders / LetsDefend / Wazuh practice	1-2 hours	Most important
Shaam	Lab writeup — aaj jo kiya uspe 5-10 sentences. GitHub commit.	20-30 min	Skip mat karo
Kabhi bhi	Darknet Diaries podcast (weekly) + English practice	20-30 min	Weekly
Weekend	GitHub writeup complete + LinkedIn post + DFIR Report / SANS article padhna	2-3 hours	Weekends focused

Phase 5 — First SOC Job Ke Baad

- Splunk + Sentinel deep dive — advanced SPL/KQL, custom dashboards, alert tuning
- Threat Hunting — hypothesis-based investigation, not just alert response
- Detection Engineering — pehla Sigma rule likho
- Incident Response — minor IR exercises lead karo
- CompTIA CySA+ — 2 saal experience ke baad shuru karo
- DFIR Report monthly padhna jaari rakho
- Estonia L2 — 3-4 saal strong India experience ke baad

COMPLETE FREE RESOURCE MASTER LIST — Reference Card

RESOURCE	URL	KYA MILTA HAI	PHASE
Cisco Skills For All	skillsforall.cisco.com	Networking + Cybersecurity path, digital badge	1-2
LinuxJourney	linuxjourney.com	Interactive Linux tutorial — best free Linux course	1
OverTheWire Bandit	overthewire.org/wargames/bandit	Hands-on Linux practice — game format, 34 levels	1
Professor Messer YouTube	YouTube: Professor Messer	Networking+ full video series — best free explanation	1-3
Microsoft Learn	learn.microsoft.com	SC-900, AZ-900, SC-200, Sentinel Ninja — all free	1-3
IBM SkillsBuild	skillsbuild.org	Cybersecurity Analyst path — IBM digital badge	1-2
Fortinet NSE 1-3	training.fortinet.com	Free cybersecurity certificates — threat landscape	1-2
subnetting.net	subnetting.net	Interactive subnetting practice	1
LetsDefend	letsdefend.io	15 SOC alerts/month + free courses: Phishing, Splunk, Linux	2-3
CyberDefenders	cyberdefenders.org	PCAP, forensics, threat hunting challenges — all free	2-3
Blue Team Labs Online	blueteamlabs.online	Weekly challenges — Splunk, Windows forensics	2
Splunk Free Training	splunk.com/en_us/training/free-courses.html	Splunk Fundamentals 1 — official free course	2
Wazuh	wazuh.com	Free SIEM+EDR platform — install and practice	2
Google Coursera	coursera.org (Financial Aid)	Google Cybersecurity Certificate — free with aid	1-2
EC-Council Essentials	eccouncil.org free courses	SOC Essentials series — free	2
MITRE ATT&CK;	attack.mitre.org	Complete adversary tactics database + Navigator	2-always
NIST SP 800-61	nvlpubs.nist.gov	Official IR standard PDF — free government document	2
SANS Reading Room	sans.org/reading-room	Free white papers — SOC, IR, threat hunting	2-3
The DFIR Report	thedfirreport.com	Real intrusion analyses — attacker TTPs	2-always
Sigma HQ GitHub	github.com/SigmaHQ/sigma	3000+ detection rules — detection logic samajhne ke liye	2-3
VirusTotal	virustotal.com	Hash/URL/IP reputation — IOC checking	2-daily
Any.Run	any.run	Interactive malware sandbox — free community tier	2-daily
CyberChef	gchq.github.io/CyberChef	Encoding/decoding/data manipulation	2-daily

TheHive Project	thehive-project.org	Case management practice	2
URLScan.io	urlscan.io	Safe URL scanning	2-daily
MxToolbox	mxtoolbox.com	Email header analysis, SPF/DKIM check	2-daily
Abuse.ch Bazaar	bazaar.abuse.ch	Malware hash lookup, botnet intelligence	2-daily
Darknet Diaries	darknetdiaries.com	Security podcast — English + mindset improvement	2-always
GDPR.eu	gdpr.eu	Official GDPR guidance — EU prep ke liye	4
regex101.com	regex101.com	Regex practice — log analysis ke liye	2

FINAL VERDICT + 12 IRON RULES

VERDICT: Yeh roadmap aapke liye SUFFICIENT hai. Isme 30 verified free resources hain. TryHackMe ko hata diya gaya hai — sab alternatives add kar diye gaye hain jo genuinely free hain. Cisco Skills For All, LinuxJourney, OverTheWire, CyberDefenders, LetsDefend, Splunk Training, Microsoft Learn — yeh sab personally verify kiye gaye hain. Agar consistently follow karo toh BCA complete hone tak aap genuinely job-ready honge. Yeh FINAL roadmap hai — ab isko mat badlo.

#	RULE	EXPLANATION
1	Roadmap dobara MAT badlo.	Yeh final hai. Naya plan dhundna band karo. Execute karo.
2	BCA ignore mat karo.	Degree + cyber balance. Ek ko sacrifice karne ki zarurat nahi.
3	Har lab ka writeup likho.	Writeup nahi = lab kiya hi nahi. Documentation = experience.
4	Roz thoda — binge nahi.	1 hour daily > 7 hours weekend. Gap aane se momentum toot ta hai.
5	Hands-on > Videos.	CyberDefenders pe actual karo. Sirf YouTube dekhne se kuch nahi hoga.
6	Certifications baad mein.	Pehle skills. SC-900 se start karo — free + relevant.
7	English daily practice karo.	Communication weak = interview fail. Chahe skill kitni bhi achi ho.
8	LinkedIn honest rakho.	Jo actually seekha wahi likho. SOC industry mein sab verify hota hai.
9	GitHub maintain karo.	Hiring manager pehle GitHub dekhta hai. Empty = no interview.
10	Wazuh zaroor install karo.	EDR skills 2026 mein important hain. Wazuh free hai — Phase 2 mein karo.
11	Investigation mindset rakho.	'Kyun?' poochho hamesha. Sirf alert close karna SOC nahi hai.
12	Labs → Writeups → Portfolio → Job.	Koi shortcut nahi. Isi sequence mein hoga.